CS664 FINAL PROJECT

Genetic Algorithm Portfolio Weights: Can Weights Trained on Old Data Survive the Modern Market?

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CS664 Artificial Intelligence · Prof. Kalathur
Boston University Metropolitan College · October 2026

Part A — Proposal

1. Motivation and Research Question

Theoretically, there is someone who wants to invest in the market and believes that old investing strategies can hold up in the modern market (60/40 portfolio). The point of this project is to test that theory, the GA will train from 2005-2019. Its weights are then frozen and kept and tested through 2020-2026. The difficulty with this is that from 2005-2019, bonds were safe whenever stocks fell, but in 2022, both stocks and bonds fell.

The question to be answered is: can the frozen weights that the GA created from the data in 2005-2019 survive the market in 2020-2026? To be able to explain the results we need the annual return, annual volatility, and the max drawdown. A GA is being used because managing 2 funds against each other is simpler than having to compare 7 at once since 7 funds have significantly more possible splits than 2, so the GA searches for a good split.

2. Data

The 7 tickers that will be used are:

- EEM (Emerging Markets Stocks)
- EFA (developed international stocks (outside the US and Canada))
- GLD (Gold)
- IWM (US small-company stocks)
- SPY (S&P 500)
- TLT (20+ year Treasury Bonds)
- VNQ (US real estate)

The data from Yahoo Finance (yfinance library) with adjusted closing prices and dates to train the GA will be from 2005-2019 and is tested against the data from 2020-2026 (Aroussi, n.d.).

3. Method

The GA will essentially create a population of portfolios, score the portfolios by their Sharpe ratio on the 2005-2019 data, then select the best scores to keep as parents. Then the GA will breed two parents and mutate the child. Each portfolio is a list of 7 weights, one per ETF, that are 0 or more and add up to 1. It is then repeated for a set number of generations where the best are kept (Russell & Norvig, 2020).

4. Evaluation Plan

The GA portfolio will then be compared to two baseline portfolios: 100% into SPY and equally splitting the capital among the 7 funds.

Those are measured by the annual return, annual volatility, Sharpe ratio, and the max drawdown. Respectively, each metric tells us how much of a return we get annually, the annual price fluctuations, the return per unit of risk, and the max drawdown is the portfolio's highest drop in percentage.

Success = if the GA has a higher Sharpe ratio than the two baselines.

- Annual return: mean(daily returns) * 252
- Annual volatility: std(daily returns) * sqrt(252)
- Sharpe ratio: (annual return - risk free rate=0)/annual volatility
- Max drawdown: the worst drop from a peak to a trough. Closer to 0 = better

5. Hypothesis

1. On 2020-2026, the GA portfolio's Sharpe ratio will be higher than the baseline of 100%SPY and equal weight.
2. The GA put most of the money into TLT, SPY, and GLD. Bonds and gold rose while stocks fell since it is more stable in 2008.
3. Something that could cause the prediction to go wrong is the year that bonds and stocks both fell instead of just stocks in 2022.

An LLM will then explain the results and reasoning behind the results.

Part B — Writeup

6. Implementation

Data

`get_data.ipynb`: Took the data for adjusted closing prices for the 7 tickers from 2005-2026 using yfinance. Printed different parts of the data pulled to check if it worked properly and was then saved as data.csv.

Helper modules

`project_helpers_metrics.py`: Created all the metrics that were being measured and used to run the GA.

`project_helpers_ga.py`: The fitness function uses the Sharpe function from project_helpers_metrics.py. It creates random portfolios, scores them with fitness, selects the best to be parents, crossover combines two parents by cutting both at a point and joining them, fitness decides whether the child is good. The mutate function mutates an ETF with a 10% chance per child to change one fund's weights then normalizes, and fixes the weights so it is greater than or equal to 0 and adds up to 1 (Ali, n.d.). The final function run_ga creates a random starting population and runs a loop for every generation. This keeps the best and returns the best weights as well as the fitness history.

Both are imported by the notebooks to run the GA as phg and phm.

Notebooks

`verification_tests.ipynb`: This tests each function to make sure that they are running as well as checking each function's output against answers worked out by hand. This notebook also brute force checks the GA finds the best possible split on a 2-fund problem (43/57) with the same Sharpe within 0.0001.

`run_ga.ipynb`: Reads data.csv and imports both phg and phm to run the simulation and creates weights based on the data. It then saves the weights as weights.csv and the best Sharpe score in each generation of the GA against the two baselines from 2005-2019 (ga_fitness_history).

`evaluate.ipynb`: After getting the weights with the training run, the weights are read and are applied to 2020-2026 where the weights are frozen and are compared against the two baselines. It then creates and saves results.csv, and the 2 figures (ga_weights and growth_test_period).

`LLM.ipynb`: Uses llama3.2 to be able to explain the results when given the data and tasked to answer the questions asked based on the weights and results (Meta, 2024; Ollama, n.d.).

Run order: Run get_data.ipynb for data, verification_tests.ipynb (to check if each function works as intended), run_ga.ipynb, evaluate.ipynb, and LLM.ipynb

7. GA Settings

Setting	Value	Reason
POP_SIZE	50	Creates a random population for the portfolios, chose 50 for a size that ran in a few seconds
N_KEEP	10	Keeps ⅕ of the population per generation. 10 parents survive as parents and are copied over unchanged. 40 children are bred each generation.
GENERATIONS	50	The generation that the fitness chart flattens at is a little over 25 generations. 50 generations was sufficient since the score stayed flat for the last 25.
RATE	0.1	There is a 10% chance that the child gets mutated, this is so it doesn't ruin the good weights that the children inherit from the parents but also tries a split the parents didn't have at a 10% chance.
SEED	6449	For reproducibility and was used throughout the class since it is my BU ID

8. Verification

Every function was tested on smaller inputs where the answer was able to be worked out by hand. For example, annual_return using 0.001, 0.002, and 0.003. The math calculated 0.002 * 252 = 0.504. The max_drawdown function test for a never falling series returned a 0 since max_drawdown calculates the most a portfolio falls from the peak to the trough. The brute force test showed that the GA found the same split as brute force (43% SPY and 57% TLT). Two funds were easier to compare with each other rather than doing all 7 since every split can be checked. These tests prove that the code calculates correctly. The brute force shows that the GA does find the real best split and proves that when the GA loses in 2020-2026, it is because of the investment strategy itself (frozen weights, mostly in bonds).

9. Results

Fund	Weight
TLT	47.3%
SPY	29.7%
GLD	16.4%
IWM	3.5%
VNQ	2.1%
EFA	0.5%
EEM	0.4%

Table 1. GA weights (trained on 2005–2019)

Portfolio	Annual return	Annual volatility	Sharpe	Max drawdown
GA	6.4%	11.4%	0.558	−26.7%
Equal weight	9.8%	15.2%	0.646	−26.0%
100% SPY	16.4%	20.0%	0.817	−33.7%

Table 2. Testing period results (2020–2026)

Portfolio	Annual return	Annual volatility	Sharpe	Max drawdown
GA	8.8%	8.3%	1.051	−16.4%
Equal weight	9.5%	16.2%	0.587	−43.0%
100% SPY	10.6%	18.7%	0.565	−55.2%

Table 3. Training period results (2005–2019)

Based on the tables above, the investment strategy from training (Table 3) the GA had the highest Sharpe ratio, but in table 2, it shows that the GA actually had the lowest Sharpe ratio during the testing period. Not only that, the GA also had the lowest return in both training and testing.

10. Figures

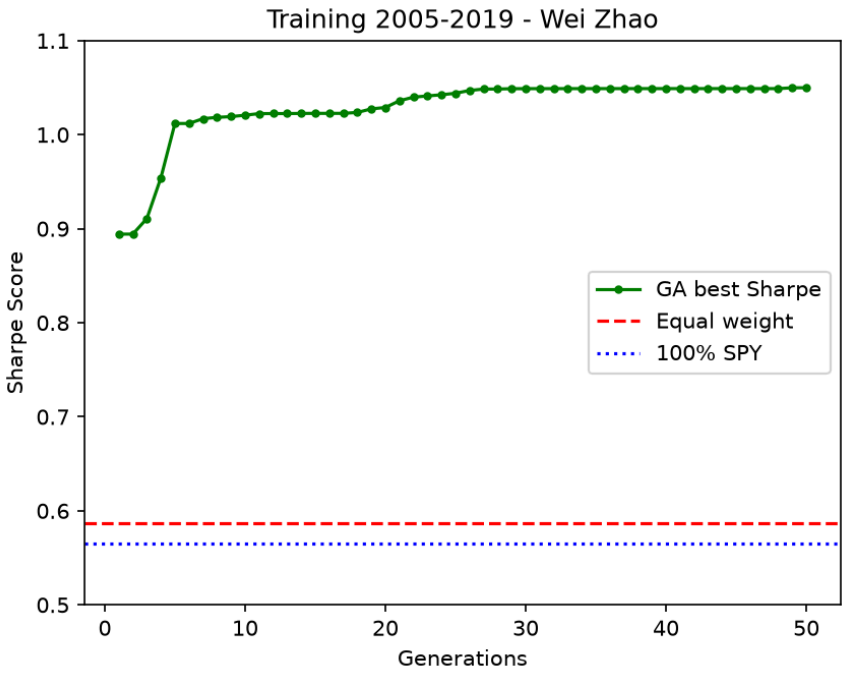


Figure 1. Shows the Sharpe score per generation. As seen, the Sharpe score flattened around generation 25. The GA's Sharpe score is seen to be far ahead of both baselines.

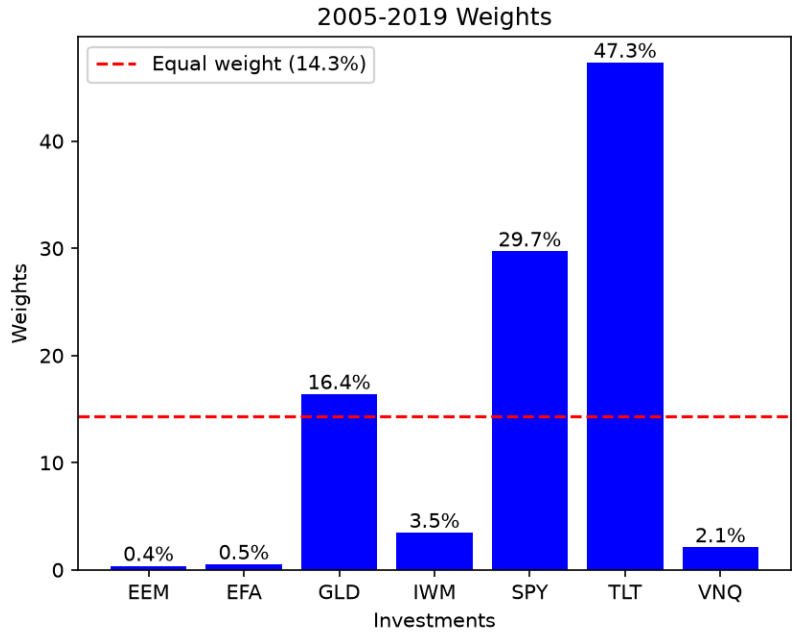


Figure 2. The most invested in were treasury bonds, then the S&P 500, then gold.

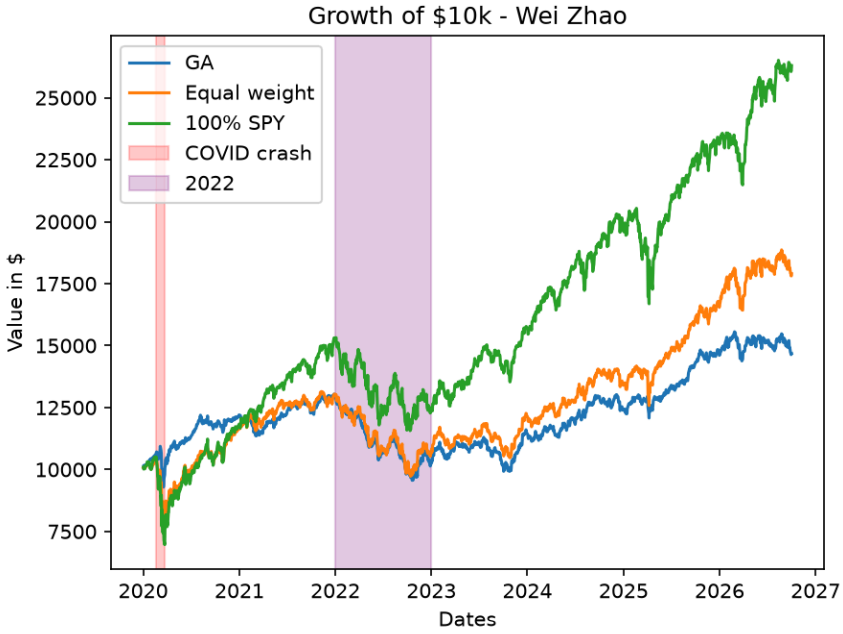


Figure 3. In red during the COVID crash, SPY and equal weight fell harder than the GA, the bonds protected the GA portfolio (Brown, 2020). They both caught up and surpassed the GA during 2021 and in 2022 both stocks and bonds fell. This caused the GA to fall behind the equal weight portfolio.

11. Results vs. Hypothesis

1. Sharpe ratio, 2020-2026 – not supported

Portfolio	Sharpe
GA	0.558
equal weight	0.646
100%SPY	0.817

2. GA weights – supported

Fund	Weight
EEM	0.4%
EFA	0.5%
GLD	16.4%
IWM	3.5%
SPY	29.7%
TLT	47.3%
VNQ	2.1%

Observed from the training output.

3. 2022 risk

From my possibilities of a cause of the experiment not working was that in 2022, both stocks and bonds fell was correct and consistent with the drop in returns compared to the 2005-2019 training run.

What held up

Even though the Sharpe was lower, the GA portfolio had the lowest volatility (11.4% vs SPY's 20%) and a smaller max drawdown (-26.7% vs -33.7%) and the risk prevention (lowest volatility) held up, but the return side did not (lowest annual return).

12. LLM Analysis

Output from llama3.2 (temperature 0, seed 6449):

> **1. Was the hypothesis supported?**
>
> No, the hypothesis was not supported. The GA portfolio's Sharpe ratio (0.558) was lower than the baseline of 100% SPY (0.817) and equal weight (0.646).
>
> **2. Why do you think the result came out this way?**
>
> There are several possible reasons why the GA portfolio's Sharpe ratio was lower than expected:
>
> • The GA portfolio's weights were optimized for the 2005-2019 period, which may not have been representative of the 2020-2026 period. The weights may have been too conservative or too aggressive for the new market conditions.
>
> • The GA portfolio's weights were held unchanged for the 2020-2026 period, which may not have allowed for sufficient adaptation to the changing market conditions. This could have resulted in suboptimal performance.

> • The GA portfolio's Sharpe ratio is sensitive to the volatility of the underlying assets. The 2020-2026 period was marked by high volatility, which may have negatively impacted the GA portfolio's Sharpe ratio.
>
> **3. What is one limitation of this experiment?**
>
> One limitation of this experiment is that it only tested the GA portfolio's performance on a single test period (2020-2026) and did not provide any information on its long-term performance. This makes it difficult to determine whether the GA portfolio's performance was a one-time anomaly or a more general phenomenon. A more comprehensive experiment would involve testing the GA portfolio's performance over multiple test periods and evaluating its long-term performance.

The model that was used was llama3.2 via Ollama. I took the weights and the results and had it analyze the data in order to answer 3 questions I asked. The LLM did get the data correct and the verdict. It also did describe the limitation of the experiment as well. Although, it said that high volatility hurt the GA, but the GA had the lowest volatility. It also never mentioned how in 2022 both stocks and bonds fell causing the GA to fall behind both baselines (reference figure 3). This is because it only received the end data and was unable to see what happened each year. The LLM was reliable at reading the data and coming up with a verdict, but not at explaining the verdict. The output has to be checked against the data before it can be trusted.

13. Discussion

In training, the GA had the highest Sharpe score. During the testing run, the GA had the lowest annual return and lowest sharpe ratio. The biggest drop was in 2022 since it had 47% in bonds and bonds fell with the stocks, the GA's safety net failed and was unable to recover as quickly as either baseline. The only thing it did better than the other portfolios was that it had the lowest volatility (11.4%). The GA only had about 30% in SPY, while SPY returned 16.4% a year from 2020 to 2026. It seems the GA traded the return for low risk. This may have worked in 2005-2019, but when the protection of low risk fails like it did in 2022, the investment strategy of investing mostly in bonds fails.

14. Limitations

There were a variety of limitations for the experiment. The LLM pointed out that the GA’s performance was only tested on a single period which was 2020-2026. The single test period was unable to determine if the 2020-2026 testing period was a typical event that happened, or if 2022 was a phenomenon that hurt the bonds. There was also 1 GA run and one seed, which means that a different seed could’ve produced different weights. There is only one random seed that creates the weight of the training and it is unknown if a different seed would have picked different weights. The test also assumes the weights are reset every day for free, which a real investor couldn’t do. Since the risk-free rate was set to 0, every portfolio’s Sharpe ratio was higher even though the ranking would’ve stayed the same. The GA also doesn’t account for trading fees either. Improvements on the

experiment could include trading fees and a real risk-free rate. Another implementation would be to run the GA with several seeds and compare the weights and to retrain the GA every few years on the most recent data, then test it on the period right after.

15. Conclusion

Back to our theoretical investor who believes that old strategies carry over to modern markets, we have seen that weights from 2005-2019 did not survive in 2020-2026. The crash of both stocks and bonds in 2022 caused the GA to fall behind the equal weight when the 100% SPY already surpassed the GA, leaving it in last place. Although the GA did not beat either baseline on Sharpe ratio, the GA had the lowest volatility. The 60/40 portfolio had its worst year since 1937 (Duberstein, 2023). The GA chose on its own a bond-heavy mix like the 60/40 portfolio that cautious investors use and was hurt the same way they were. This means that an old strategy can fail when the pattern it relies on stops being true. In this experiment, the pattern was bonds rising when stocks fall.

References

Ali, R. (n.d.). *Genetic algorithm with Python made easy: Code + easy explanation*. Medium. https://medium.com/@Data_Aficionado_1083/genetic-algorithm-with-python-made-easy-code-easy-explanation-87c3ad6ca152

Aroussi, R. (n.d.). *yfinance* (Version 1.7.0) [Python library]. https://github.com/ranaroussi/yfinance

Brown, C. (2020, August 18). *S&P 500 closes at record high for first time since pandemic began*. Axios. https://www.axios.com/2020/08/18/sp500-stock-record-high-pandemic

Duberstein, B. (2023, January 26). *2022 was the worst year since 1937 for this investing strategy*. The Motley Fool. https://www.fool.com/investing/2023/01/26/2022-was-the-worst-year-since-1937-for-this-invest

Kalathur, S. (2026). *Ollama Part 1* [Course notes]. CS664 Artificial Intelligence, Boston University Metropolitan College.

Matplotlib Development Team. (n.d.). *matplotlib.axes.Axes.bar_label*. https://matplotlib.org/stable/api/_as_gen/matplotlib.axes.Axes.bar_label.html

Meta. (2024). *Llama 3.2* [Large language model]. https://ollama.com/library/llama3.2

Ollama. (n.d.). *Ollama* [Computer software]. https://ollama.com

Python Software Foundation. (n.d.). *Format specification mini-language*. https://docs.python.org/3/library/string.html#format-specification-mini-language

Russell, S., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Yahoo Finance. (n.d.). *Historical price data for EEM, EFA, GLD, IWM, SPY, TLT, VNQ* [Data set]. Retrieved October 2026, from https://finance.yahoo.com

Acknowledgments

Thanks to my brother for help writing the brute-force verification loop.